

$$\begin{aligned}
(22a) \rightarrow & \frac{1}{16\pi^2} \left(\frac{1}{9} g_2^4 \text{LF}_{3,0}[\mathbf{m}_2] + \frac{1}{6} g_2^4 \text{LF}_{4,-1}[\mathbf{m}_2] - \frac{8}{45} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_2] - \right. \\
& \frac{1}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[\mathbf{m}_d^{\mathbf{p}}] + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_d^{\mathbf{p}}] - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[\mathbf{m}_d^{\mathbf{p}}] - \frac{1}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[\mathbf{m}_e^{\mathbf{p}}] + \\
& \frac{5}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^{\mathbf{p}}] - \frac{4}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[\mathbf{m}_e^{\mathbf{p}}] + \frac{1}{18} \sum_{\mathbf{p}} (-g_1^4 + g_2^4) \text{LF}_{3,0}[\mathbf{m}_l^{\mathbf{p}}] + \\
& \frac{5}{48} \sum_{\mathbf{p}} (g_1^4 - g_2^4) \text{LF}_{4,-1}[\mathbf{m}_l^{\mathbf{p}}] - \frac{2}{45} \sum_{\mathbf{p}} (g_1^4 - g_2^4) \text{LF}_{5,-2}[\mathbf{m}_l^{\mathbf{p}}] - \frac{1}{54} \sum_{\mathbf{p}} (g_1^4 - 9g_2^4) \text{LF}_{3,0}[\mathbf{m}_q^{\mathbf{p}}] + \\
& \frac{5}{144} \sum_{\mathbf{p}} (g_1^4 - 9g_2^4) \text{LF}_{4,-1}[\mathbf{m}_q^{\mathbf{p}}] - \frac{2}{135} \sum_{\mathbf{p}} (g_1^4 - 9g_2^4) \text{LF}_{5,-2}[\mathbf{m}_q^{\mathbf{p}}] - \\
& \frac{1}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[\mathbf{m}_u^{\mathbf{p}}] + \frac{5}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_u^{\mathbf{p}}] - \frac{16}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[\mathbf{m}_u^{\mathbf{p}}] + \\
& \frac{1}{18} (-g_1^4 + g_2^4) \text{LF}_{3,0}[\tilde{\mu}] + \frac{1}{12} (-g_1^4 + g_2^4) \text{LF}_{4,-1}[\tilde{\mu}] + \frac{4}{45} (g_1^4 - g_2^4) \text{LF}_{5,-2}[\tilde{\mu}] + \\
& \frac{1}{16} g_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \tilde{\mu}] + \frac{1}{8} g_1^4 m_1^2 \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] - \frac{2}{3} g_2^4 \text{LF}_{2,1,0}[\mathbf{m}_2, \tilde{\mu}] - \\
& \frac{17}{48} g_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{3} g_2^4 \text{LF}_{3,1,-1}[\mathbf{m}_2, \tilde{\mu}] + \\
& \frac{1}{3} g_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{6} g_1^2 (\overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} + \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}}) \text{LF}_{2,2,0}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{12} g_1^2 (\overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} + \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \frac{1}{6} g_1^2 (\overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} + \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}}) \\
& \text{LF}_{2,2,0}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}] + \frac{1}{12} g_1^2 (\overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} + \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}] + \\
& \frac{1}{6} g_1^2 (\overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} + \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \frac{1}{6} g_1^2 (\overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} + \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}}) \\
& \text{LF}_{3,2,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] - \frac{1}{8} (3g_1^2 - g_2^2) (\overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} + \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \\
& \frac{1}{6} (g_1^2 - g_2^2) (\overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} + \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \frac{1}{6} g_1^2 (\overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} + \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}}) \\
& \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \frac{1}{6} g_1^2 (\overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} + \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] - \\
& \frac{1}{8} (5g_1^2 - 3g_2^2) (\overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} + \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{1}{2} (g_1^2 - g_2^2) (\overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} + \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} + \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}}) \text{LF}_{2,2,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \\
& \frac{1}{24} (g_1^2 + 3g_2^2) (\overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} + \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \\
& \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} + \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} + \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{1}{4} (g_1^2 - 3g_2^2) (\overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} + \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{2} (g_1^2 - g_2^2) (\overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} + \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{3}{2} \overline{y}_d^{\text{pr}} y_d^{\text{sr}} \overline{y}_u^{\text{st}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] - \frac{3}{2} \overline{y}_d^{\text{pr}} y_d^{\text{sr}} \overline{y}_u^{\text{st}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{1}{8} g_1^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{8} g_1^4 m_1^2 \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] + \frac{5}{12} (g_1^4 - g_1^2 g_2^2) \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] - \\
& \frac{1}{8} g_1^4 \text{LF}_{4,2,-3}[\tilde{\mu}, \mathbf{m}_1] + \frac{1}{8} g_1^4 m_1^2 \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \frac{1}{3} (-g_1^4 + g_1^2 g_2^2) \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] + \\
& \frac{4}{3} g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] + \frac{31}{24} g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_2] - \frac{1}{8} g_2^4 m_2^2 \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] + \\
& \frac{1}{4} g_2^2 (5g_1^2 -$$